

**FACULDADE DE ENGENHARIA DA UNIVERSIDADE DO PORTO**

**<Dissertation Title>**

**<Author's Full Name>**

**VERSÃO DE TRABALHO**



Mestrado em Engenharia Eletrotécnica e de Computadores

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12 de janeiro de 2026



# Resumo

(Por fazer)

# Abstract

(TODO)

# UN Sustainable Development Goals

The United Nations Sustainable Development Goals (SDGs) define a global framework for promoting sustainable development through social, economic, and technological progress. Although this thesis is not directly focused on addressing a specific SDG, it contributes indirectly by developing a low-cost technological solution that supports accessibility and innovation in sports training environments.

In particular, this work is loosely aligned with the following goal:

**SDG 9 – Industry, Innovation and Infrastructure:** The development of a low-cost, computer-vision-based automatic scoring system demonstrates the application of digital technologies and innovation to improve existing infrastructures in sports facilities, making advanced training tools more accessible to amateur clubs and individual athletes.

# Acknowledgements

(TODO)

<Christian Geyer>

*“Our greatest glory is not in never falling, but in rising every time we fall”*

Confucius

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# List of Acronyms

|      |                                          |
|------|------------------------------------------|
| CNN  | Convolutional Neural Network             |
| ISSF | International Shooting Sports Federation |

# Capítulo 1

## Introduction

(TODO)

## Capítulo 2

# Problem Definition and State of the Art

### 2.1 Introduction

Scoring targets is an essential task in sport shooting, and a variety of systems have been developed over the years to replace the time-consuming manual scoring. Common technologies include triangulation of acoustic signals, laser-beam interruption and impact sensing. These technologies represent high-accuracy but high-cost solutions, and are well-suited for professional events. Many amateur sport-shooting clubs are interested in low-cost, real-time scoring systems that do not require the same level of precision. Computer-vision-based systems offer a low-cost alternative capable of detecting and scoring paper targets.

This section reviews the state of the art in computer-vision-based target scoring systems. This includes classical image processing methods, and more recent deep learning approaches.

### 2.2 Physical and Regulatory Requirements of ISSF Paper Targets

ISSF defines the dimensions and scoring rules for paper targets used in 10 m air rifle and 10 m air pistol events.

Tabela 2.1: ISSF 10 m Air Rifle target ring diameters [1]

| Ring | Diameter (mm) |
|------|---------------|
| 10   | 0.5           |
| 9    | 5.5           |
| 8    | 10.5          |
| 7    | 15.5          |
| 6    | 20.5          |
| 5    | 25.5          |
| 4    | 30.5          |
| 3    | 35.5          |
| 2    | 40.5          |
| 1    | 45.5          |

Tabela 2.2: ISSF 10 m Air Pistol target ring diameters [1]

| Ring | Diameter (mm) |
|------|---------------|
| 10   | 11.5          |
| 9    | 27.5          |
| 8    | 43.5          |
| 7    | 59.5          |
| 6    | 75.5          |
| 5    | 91.5          |
| 4    | 107.5         |
| 3    | 123.5         |
| 2    | 139.5         |
| 1    | 155.5         |



Figura 2.1: ISSF 10 m Air Rifle target [1]



Figura 2.2: ISSF 10 m Air Pistol target [1]

Figure 2.1 illustrates the ISSF 10 m air rifle paper target, while Fig. 2.2 shows the corresponding 10 m air pistol target. The official ring diameters for both targets are listed in Table 2.1 and Table 2.2, respectively.

For the 10 m air rifle paper target, the innermost ring (ring 10) is reduced to a small white dot at the center of the target with a diameter of 0.5 mm, as shown in Fig. 2.1. The ring diameters increase arithmetically by 5 mm up to the outermost ring (ring 1), which has a diameter of 45.5 mm, as summarized in Table 2.1. The inner black region extends from ring 10 to ring 4, with a total diameter of 30.5 mm.

The 10 m air pistol paper target is larger, as shown in Fig. 2.2. The innermost ring (ring 10) has a diameter of 11.5 mm, while the outermost ring (ring 1) has a diameter of 155.5 mm. The ring diameters increase arithmetically by 16 mm, as listed in Table 2.2. The inner black region extends from ring 10 to ring 7, with a diameter of 59.5 mm.

The dimensions summarized in Fig. 2.1 and Table 2.1 highlight the extremely small scale of the air rifle target, which directly motivates the accuracy analysis presented in the following section.

The continuous radial scoring model can be written as

$$s_c(r) = 11.0 - kr, \quad (2.1)$$

where  $r$  is the radial distance between the shot center and the target center, and  $k$  is a scale factor and corresponds to the inverse radius increment

$$k = \frac{2}{\Delta d} \quad (2.2)$$

between adjacent rings.

ISSF decimal scoring applies rounding and clipping to the continuous scoring model with the

formula

$$s(r) = \text{clip}\left(\frac{\lfloor 10s_c(r) \rfloor}{10}, 0, 10.9\right), \quad (2.3)$$

where  $\lfloor \cdot \rfloor$  denotes the floor operator and  $\text{clip}(x, a, b) = \min(\max(x, a), b)$  limits the score to the valid ISSF range.

## 2.3 Accuracy Requirements

The scoring scale resolution and the smallest paper target size impose accuracy requirements for any scoring system. Due to the small dimension of the 10m-air-rifle paper target, the required accuracy goes into the sub millimeter range.

To understand the accuracy requirements, we analyse the measurements relative to one interval of constant score. The length of the interval is given by

$$\Delta r_{0.1} = 0.2 \cdot \Delta d \quad (2.4)$$

and corresponds to the increment in radial distance that produces a change of 0.1 in the continuous score.

The measurement error is modeled as a uniform random variable in the interval  $[-E, E]$ . Let  $R$  denote the ground-truth radial position within a decimal scoring interval and let  $\varepsilon$  denote the measurement error. The measured value is  $M = R + \varepsilon$ . Assuming  $R$  and  $\varepsilon$  are independent, the probability density function (pdf) of  $M$  is given by the convolution of the pdfs of  $R$  and  $\varepsilon$ , as illustrated in Fig. 2.3.

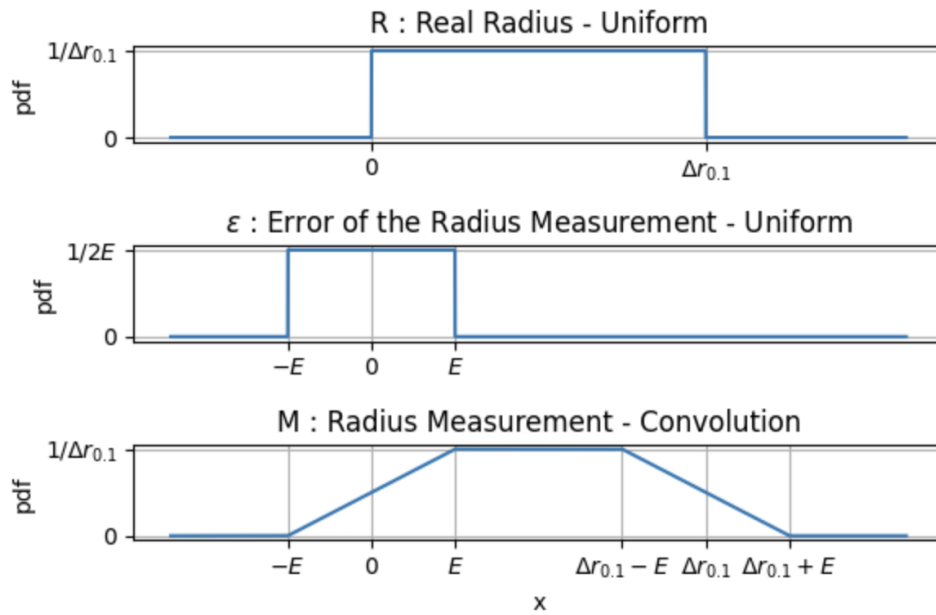


Figure 2.3: Bounded-error model: uniform ground-truth position within a scoring interval and uniform measurement error, and the resulting measurement pdf obtained by convolution.

Regardless of how small the error is, there will always be a non-zero probability for incorrect scoring by 0.1. The probability of correctly scoring a bullet is given by the integral of the pdf of the measurement within the scoring interval, which is

$$P = 1 - \frac{E}{2 * \Delta r_{0.1}} \quad (2.5)$$

.

Solving for the accuracy  $E$ , we obtain:

$$E = 2\Delta r_{0.1} (1 - P) \quad (2.6)$$

.

Tabela 2.3: Required measurement accuracy  $E$  as a function of the probability  $P$  of correct decimal scoring for a 10 m air rifle target ( $\Delta r_{0.1} = 0.25$  mm).

| Probability $P$ (%) | Required accuracy $E$ (mm) |
|---------------------|----------------------------|
| 80                  | 0.10                       |
| 85                  | 0.075                      |
| 90                  | 0.050                      |
| 95                  | 0.025                      |
| 99                  | 0.005                      |

Table 2.3 shows that achieving high probabilities of correct decimal scoring requires increasingly strict bounds on the radial measurement error. For the 10 m air rifle target, a probability of 90% already requires a measurement accuracy better than 0.05 mm, while probabilities above 95% demand accuracies below 0.03 mm. These values illustrate the high accuracy requirements imposed by ISSF decimal scoring and motivate the use of precise geometric localization methods in automatic paper target scoring systems.

## 2.4 Classical Computer Vision Approaches

Classical computer vision approaches to paper target scoring typically rely on explicit image segmentation and geometric reasoning. In contrast to learning-based methods, they aim to recover physically meaningful quantities—such as the target center, ring radii and bullet-hole centers—and then compute the score by applying the official scoring rules. These pipelines can achieve high localization accuracy under controlled imaging conditions, but they often become fragile when lighting varies, the background is cluttered, or bullet holes overlap.

### 2.4.1 Geometric scoring with perspective correction and contour-based circle fitting

A representative geometric pipeline is proposed by Yu *et al.* [4], who score paper targets using only classical image processing and circle fitting. The method first computes a projective trans-



formation to obtain a fronto-parallel view of the target, compensating for camera viewpoint. After perspective correction, the outer contour of the target is extracted and a circle is fitted to estimate the target center and scale. Bullet holes are then isolated via thresholding and morphological operations, and connected components are extracted. For each bullet-hole region, contours are computed and a circle model (e.g., minimum enclosing circle or least-squares circle fitting) is fitted to estimate the bullet-hole center. Finally, the score is computed from the distance between the bullet-hole center and the target center using the known ISSF ring geometry.

The main strength of this approach is that it explicitly reconstructs the target geometry and estimates centers via contour-based fitting, which is well-aligned with ISSF-compliant scoring. However, it assumes that segmentation is reliable, which is typically achieved under controlled backgrounds and stable lighting. As a result, performance degrades when bullet holes are poorly contrasted against the paper, when shadows or glare appear, or when bullet holes overlap, since the connected components no longer correspond to individual impacts.

#### **2.4.2 Segmentation of overlapping bullet holes using watershed and circle fitting**

Overlapping impacts (“heavy holes”) are a major failure mode for connected-component-based pipelines. Liu *et al.* [3] address this scenario using an improved watershed segmentation strategy. Their pipeline applies grayscale preprocessing and morphological operations, followed by area-based filtering to suppress non-hole regions such as ring markings and text. Overlapping bullet holes are then separated using a distance-transform-driven watershed method, and boundaries are extracted (e.g., via chain-code tracking). For each separated region, least-squares circle fitting is used to estimate individual hole centers and radii.

This work demonstrates that, when the preprocessing is successful, classical segmentation can separate strongly overlapping holes and still recover geometrically meaningful centers. Nonetheless, the method depends on threshold choices and assumptions about bullet-hole appearance (e.g., expected region areas). In addition, the paper focuses on cropped regions around overlapping holes rather than full target images, so it does not fully address end-to-end robustness in a complete target scoring system.

#### **2.4.3 Temporal differencing and Hough-based circle detection for live scoring**

A complementary classical strategy for live scoring is to exploit temporal information. Li *et al.* [2] propose a method that processes pairs of images captured before and after each shot. After geometric correction (homography), the pre-shot image is subtracted from the post-shot image, suppressing static content such as rings, previous holes and background structure. The resulting difference image highlights only the newly created bullet hole. The pipeline then applies binarization, morphological cleaning and edge detection, followed by a generalized Hough transform to detect the circular bullet-hole contour and estimate its center. A Hough transform can also be used to localize a reference ring or the target center, enabling score computation from the measured radial distance.

The main advantage of temporal differencing is robustness to cluttered targets, since it avoids the need to segment all existing holes simultaneously. However, it requires a stable camera and assumes that exactly one new shot occurs between consecutive frames; camera motion or multiple new impacts can break the subtraction assumption. Despite these constraints, temporal differencing is attractive as an optional component in real-time systems where images are captured continuously and the shot timing is known.

#### 2.4.4 Discussion

Overall, classical methods are attractive because they compute scores from explicit target geometry and can be computationally lightweight. Their limitations arise primarily from the sensitivity of segmentation to illumination changes, background clutter, target mounting frames and heavy target wear. In addition, classical pipelines are typically parameter-heavy, relying on multiple thresholds and heuristic parameters (e.g., intensity thresholds, morphological kernel sizes and area constraints) that must be carefully tuned for each imaging setup. These observations motivate hybrid designs that preserve geometric scoring and center estimation, while improving robustness through stronger detection mechanisms (e.g., learning-based detectors or temporal differencing when available).

## 2.5 Deep Learning-Based Approaches

Recent work has explored the use of deep learning, particularly convolutional neural networks (CNNs), for automatic detection of bullet holes on paper targets. These approaches typically frame the problem as an object detection or instance segmentation task, with the goal of localizing bullet holes under challenging imaging conditions.

A representative study is presented by Butt *et al.* [butt2024\_yolov8], who apply the YOLOv8 object detection framework to detect bullet holes on indoor shooting targets. The authors report high detection performance, with mean Average Precision (mAP) values exceeding 95% under controlled indoor conditions. A key advantage of the YOLOv8-based approach is its computational efficiency: inference times on consumer-grade GPUs are on the order of a few milliseconds per image, making the method suitable for real-time or near-real-time applications. The study demonstrates that CNN-based detectors can robustly identify bullet holes despite moderate lighting variations and background clutter, outperforming classical threshold-based methods in detection reliability.

However, the detected bullet holes are represented by bounding boxes, and the paper does not address sub-region analysis or precise center estimation beyond the detector output. Consequently, while detection accuracy is high, the geometric precision required for ISSF-compliant decimal scoring is not directly achieved.

A related line of work is presented by Ahmed *et al.* [ahmed2023\_maskrcnn], who employ a Mask R-CNN architecture with a ResNet-50 backbone to detect and segment bullet holes on outdoor shooting targets. This work demonstrates the robustness of deep learning methods in

more adverse conditions, including variable illumination, weather effects and non-uniform backgrounds. The use of instance segmentation allows the network to recover the approximate shape of bullet holes rather than only bounding boxes, which is advantageous for visual interpretation and robustness.

Despite this increased robustness, the reported localization accuracy remains limited by the resolution of the segmentation masks and the network output grid. The study focuses primarily on detection and segmentation performance metrics and does not explicitly analyze the accuracy of the estimated bullet-hole centers relative to the precision required for ISSF decimal scoring. In addition, overlapping bullet holes are not explicitly addressed, and the segmentation results are evaluated mainly on isolated impacts.

Overall, deep learning approaches offer significant advantages in terms of robustness to illumination changes, background clutter and target wear, and they enable fast, real-time detection on low-cost hardware. However, accurately estimating bullet-hole center coordinates with sub-millimeter precision remains a major challenge. The detector outputs (bounding boxes or coarse masks) are not sufficiently precise for high-accuracy scoring on their own, and none of the reviewed deep learning studies explicitly address the problem of overlapping bullet holes. These limitations motivate hybrid approaches that combine CNN-based detection for robustness with classical geometric refinement for precision.

## 2.6 Discussion and Summary

This chapter reviewed the physical and regulatory constraints imposed by ISSF paper targets and analyzed their implications for automatic scoring systems. Due to the small dimensions of the 10 m air rifle target and the use of decimal scoring, even small localization errors can lead to incorrect scores. A simple bounded-error model was introduced to illustrate how scoring accuracy depends directly on the measurement error, highlighting the high accuracy requirements imposed by ISSF rules.

The state of the art in computer-vision-based target scoring was then reviewed. Classical image processing approaches rely on explicit geometric modeling and can achieve high localization accuracy under controlled conditions. However, these methods are sensitive to illumination changes, background clutter, target wear and overlapping bullet holes, and they often require careful tuning of multiple heuristic parameters. In contrast, deep learning-based approaches demonstrate strong robustness and real-time performance, particularly for bullet-hole detection, but their outputs are typically limited to bounding boxes or coarse segmentation masks, which do not provide the geometric precision required for ISSF-compliant decimal scoring.

Overall, the literature shows that no single approach fully satisfies both the robustness and precision requirements of automatic ISSF paper target scoring. These observations motivate a hybrid system design that combines the robustness of learning-based detection with the precision of classical geometric refinement. The following chapter presents the proposed system architecture based on this hybrid approach.

## Capítulo 3

# Proposed System Architecture

### 3.1 System Overview

The proposed system is designed to provide automatic scoring of ISSF-compliant paper targets using low-cost imaging hardware and computer vision techniques. In addition to shot detection and scoring, the system supports data storage and visualization to enable the management of athletes, training sessions and competitions.

At a high level, the system consists of five main components: image acquisition, image processing, scoring logic, data management and visualization. A fixed USB camera captures images of the paper target after each shot and transmits them to a personal computer for processing. The image-processing subsystem analyzes the acquired images to localize the target, detect bullet holes and estimate their geometric properties. Based on these measurements, the scoring module computes ISSF-compliant decimal scores. The resulting shot data and scores are stored in a database, where they are associated with athletes, sessions and competitions, and can be visualized through a virtual target representation.

Figure 3.1 illustrates the high-level architecture of the proposed system and the main processing stages from image acquisition to score visualization.

While the complete system provides end-to-end functionality for automatic target scoring, the primary focus of this thesis is the image-processing subsystem. In particular, the research addresses the problem of robust and accurate bullet-hole localization under the constraints identified in Chapter 2. The remaining components, including data storage and visualization, serve to integrate the image-processing results into a usable scoring system but are not the main object of investigation.

The image-processing subsystem is designed as a modular pipeline, allowing different detection and refinement strategies to be evaluated and compared. Robust bullet-hole detection is treated separately from precise geometric refinement, enabling the system to combine complementary approaches and to balance robustness and accuracy. The following sections describe the design of this subsystem in detail, beginning with image acquisition assumptions and progressing through detection, refinement and scoring.

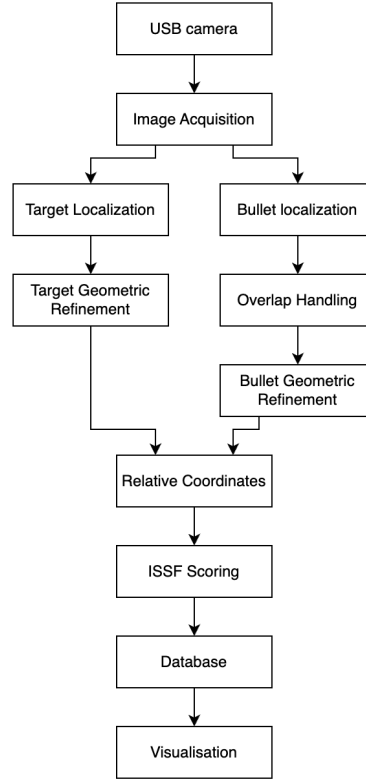


Figure 3.1: High-level architecture of the proposed system.

### 3.2 Image Acquisition and Operating Assumptions

Images of the paper target are acquired using a fixed USB camera connected to a personal computer. The camera is mounted at a known and approximately constant distance from the target and remains stationary during operation. This fixed setup allows the use of geometric constraints and calibration procedures in later processing stages.

The system is intended to operate in an indoor shooting range, where lighting conditions can be controlled to a certain extent but are not assumed to be perfectly uniform. Variations in illumination, shadows and reflections may still occur, particularly due to the presence of a metal target frame and wear of the paper target over time. No additional white background board is assumed behind the target by default.

However, the operating environment allows for limited control of background appearance if required. In particular, the use of a uniform background color behind the target can be introduced as a mitigation measure in scenarios where segmentation-based methods are adversely affected by background clutter or low contrast. This option provides additional flexibility while remaining consistent with the low-cost and controlled indoor deployment scenario considered in this work.

Images are captured at a sufficient resolution to allow reliable detection of bullet holes and target features. Image acquisition is triggered after each shot, either manually or through an external signal, such that each image contains at most one new bullet hole relative to the previous state of the target.

These assumptions reflect a realistic low-cost deployment scenario and are consistent with the constraints discussed in Chapter 2. They guide the design of the image-processing pipeline described in the following sections.

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## **Anexo A**

### **Some chapter**

(TODO)